

MEMBER 5: INTEGRATION & SIH LEAD - VIBE CODING ROADMAP

Smart India Hackathon 2026 | Team ILLUMINATI (ID: 27113) | PS: SIH1642

Primary Folder: tests/, docs/ | Dedicated Branch: feature/integration-docs | Deliverables: 4 Files

PHASE 1: ML PIPELINE AUTOMATED TESTS

Prompt 5.1

Create tests/test_ml_pipeline.py. Write test_baseline_calculation() asserting that mu, sigma, and IQR are non-null and valid for all 10 cities across all months.

Prompt 5.2

In tests/test_ml_pipeline.py, write test_physical_bounds_validator() feeding impossible inputs (Humidity=500%, Temp=-999 deg C) and asserting they are correctly filtered out.

Prompt 5.3

In tests/test_ml_pipeline.py, write test_isolation_forest_score_bounds() running 100 random weather vectors and asserting all anomaly scores lie strictly within [0.00, 1.00].

Prompt 5.4

In tests/test_ml_pipeline.py, write test_normal_weather_scenario() providing typical September Bengaluru weather (26 deg C, 10mm rain, 70% humidity) and asserting severity is NORMAL and score < 0.40.

Prompt 5.5

In tests/test_ml_pipeline.py, write test_extreme_cloudburst_scenario() providing 37 deg C, 145mm rain, 92% humidity, 994 hPa and asserting severity is CRITICAL (score >= 0.90) with rainfall as #1 contributor.

Prompt 5.6

In tests/test_ml_pipeline.py, write test_explainability_percentages() asserting the sum of all contributor percentages equals 100.0% +/- 0.5%.

PHASE 2: BACKEND API INTEGRATION TESTS

Prompt 5.7

Create tests/test_backend_api.py using FastAPI TestClient (httpx). Write test_health_endpoint() asserting GET /health returns 200 OK and status 'ok'.

Prompt 5.8

In tests/test_backend_api.py, write test_locations_endpoint() asserting GET /locations returns 10 cities with lat, lon, and active severity.

Prompt 5.9

In tests/test_backend_api.py, write test_predict_endpoint_valid() asserting POST /predict returns valid schema with score, severity, contributors, and explanation.

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Prompt 5.10

In tests/test_backend_api.py, write test_predict_endpoint_invalid() sending negative humidity (-20%) and asserting HTTP 400 or 422 with actionable error message.

Prompt 5.11

In tests/test_backend_api.py, write test_history_endpoint() asserting GET /history/Bengaluru returns time-series data points with observed and expected values.

PHASE 3: BENCHMARK MODEL EVALUATION & METRICS

Prompt 5.12

Create eval/evaluate_models.py ingesting test dataset with labeled ground-truth extreme events from Member 2.

Prompt 5.13

In eval/evaluate_models.py, implement baseline comparison models: 1. Static Threshold Model (temp > 35 or rain > 50), 2. Standalone Z-Score Model, 3. Standalone Isolation Forest, 4. Proposed Dual-Engine Model.

Prompt 5.14

In eval/evaluate_models.py, compute standard classification metrics for all 4 models: Precision, Recall, F1-Score, and ROC-AUC.

Prompt 5.15

In eval/evaluate_models.py, measure inference latency across 1,000 predictions and calculate 95th percentile latency (target: <15ms).

Prompt 5.16

Generate formatted Markdown evaluation report and save to docs/evaluation_results.md containing comparison tables and metric graphs.

PHASE 4: SIH PRESENTATION DRILL & DEMO RUNBOOK

Prompt 5.17

Create docs/demo_runbook.md writing a minute-by-minute live demonstration checklist matching our 6-slide presentation (docs/presentation_content.md).

Prompt 5.18

In docs/demo_runbook.md, detail exact click sequence: 1. Show National Anomaly Map, 2. Click Bengaluru pin to reveal deep-dive dials, 3. Load Bengaluru Cloudburst preset in What-If Studio, 4. Point out 0.94 CRITICAL score and explainability breakdown, 5. Show historical trend chart.

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Prompt 5.19

In docs/demo_runbook.md, add 30-second speaking scripts for each member during the live judging demo so all 5 members speak.

Prompt 5.20

Conduct end-to-end integration dry run: verify uvicorn backend.main:app and npm run dev run with zero console errors. Log final sign-off in memory.md.